

Midterm Project Report for AI-Driven Fashion Production

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Abstract

This project aims to tackle zero-waste fashion production through artificial intelligence. We focus on optimizing pattern piece layout using reinforcement learning to minimize fabric waste, and automating tech pack generation with multimodal AI models. Our work addresses the estimated 15% fabric waste in garment production, proposing computational solutions to environmental and efficiency concerns in the fashion industry.

1 Summary

This project aims to tackle the challenge of zero-waste fashion production through artificial intelligence techniques. We focus on two key aspects: optimizing the layout of pattern pieces to minimize fabric waste using reinforcement learning, and automating tech pack generation through multimodal AI models. Our work addresses the estimated 15% of fabric typically wasted during garment production, proposing novel computational solutions to environmental and efficiency concerns in the fashion industry.

In the initial stages of our project, we conducted extensive research analyzing the state-of-the-art in reinforcement learning for 2D nesting problems and multimodal models for technical documentation generation. By examining key research including deep reinforcement learning for rectangular strip packing (Fang et al., 2023), vision-based bin packing (Kundu et al., 2019), and

multimodal LLMs like GPT-4 Vision and LLaVA, we established a solid theoretical foundation for our approach.

Through our literature review and analysis, we have developed a comprehensive roadmap for implementation. We have carefully compared different methodologies and algorithms to determine the most promising approaches for our specific use case. Additionally, we have conducted initial experiments with the LLaVA model to explore its capabilities for component identification in garment sketches.

Our current work focuses on exploring sketch-to-vector conversion possibilities and evaluating different approaches for component recognition. By mapping out the complete pipeline from design sketches to optimized patterns and tech pack generation, we aim to create an integrated system that supports sustainable fashion production through AI-driven optimization and automation.

2 Literature Review

2.1 Layout Optimization with Reinforcement Learning

One critical aspect of zero-waste fashion is optimizing the layout of pattern pieces to minimize fabric waste. Recent research in reinforcement learning (RL) and other AI techniques has tackled this 2D nesting problem in manufacturing with promising results.

Fang et al. (2023) proposed a novel deep RL approach for 2D packing (strip nesting) using a pointer-network encoder-decoder policy and model-free RL to decide placement sequences without hand-crafted heuristics. A bottom-left packing algorithm is then applied to place pieces, with rewards guiding the RL agent. This method achieved comparable or better material utilization than classical heuristics, with fast solution times (less than 1 second) and good generalization to new instances. While focused on rectangular shapes rather than the irregular patterns common in fashion, the approach offers valuable insights for weight coefficient evaluation and hybrid optimization techniques.

Addressing the limitations of rectangular packing, Fang et al. (2023) also introduced one of the first RL algorithms specifically for irregularly-shaped pattern packing in a separate study. Their approach combines Monte Carlo learning, Q-learning, and Sarsa in a hybrid RL scheme tailored to irregular pieces. The Proximal Policy Optimization (PPO) algorithm effectively handles irregular shape nesting, achieving 5-8% better material utilization than traditional methods. Their custom action space design enables precise rotation and placement decisions essential for garment pattern optimization, while their state representation technique successfully captures partially-filled layouts, making it adaptable to the complex spatial relationships in fashion pattern cutting.

Kundu et al. (2019) developed Deep-Pack, a pioneering method that treats the packing layout as an image and uses deep reinforcement learning to place items. A convolutional neural network observes the bin canvas, and an RL agent (trained via deep Q-learning) decides where to place the next piece. This vision-based strategy enables handling arbitrary item arrangements in an online 2D bin packing setting. The approach uses a Pointer Network architecture and policy gradient methods to learn place-

ment policies for irregular shapes, achieving near-optimal solutions without explicit programming of heuristics.

Pugliese et al. (2025) built a custom OpenAI Gym environment for a constrained 2D packing problem, employing Proximal Policy Optimization from Stable Baselines3 to train agents for piece placement. Their groundbreaking advancement improved the PPO algorithm by incorporating a novel adaptive clipping mechanism that dynamically adjusts the clipping parameter based on pattern complexity. OpenAI Gym, a toolkit for developing and comparing reinforcement learning algorithms, provides standardized environments with common interfaces, making it ideal for our application.

As shown in Figure 1, their approach allows for dynamic adaptation to pattern complexity. Results showed the learned policy outperforming classic heuristics in material utilization. Their two-stage approach combining deep reinforcement learning with mathematical programming achieves superior material utilization compared to traditional algorithms while maintaining computational efficiency—crucial for real-time fashion design applications.

2.2 Tech Pack Generation with Multimodal Models

Another component of our project is generating tech packs—the technical specifications and documentation for garments—automatically from design inputs. Recent advances in multimodal large language models offer significant potential in this area.

OpenAI’s GPT-4 introduced multimodal capabilities allowing it to interpret images and generate detailed text. A famous demonstration showed GPT-4 transforming a hand-drawn webpage sketch into working HTML/CSS code, suggesting that a vision-enabled LLM can produce structured outputs based on visual understanding. By analogy, this capability can be applied to

The mathematical formulation for their PPO implementation can be expressed as:

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t) \right] \quad (1)$$

Where $r_t(\theta)$ represents the ratio of the probability under the new policy to the probability under the old policy, and $\hat{A}_t$ is the advantage estimation at time t . The clipping function restricts the policy update to a small neighborhood around the current policy, preventing excessively large updates that could destabilize training. Their innovation was to make ϵ dynamic:

$$\epsilon_t = \epsilon_{\text{base}} \cdot \alpha^{c_t} \quad (2)$$

Where c_t represents the complexity measure of the current pattern layout and α is a scaling factor.

Figure 1: PPO Algorithm with Adaptive Clipping for Pattern Layout Optimization

fashion: feeding an image of a garment sketch or pattern layout into GPT-4 could generate a structured tech pack including measurements, bill of materials, and sewing instructions.

Pothier (2024) demonstrated a practical workflow using GPT-4’s vision API to extract structured data from images. By integrating an LLM with frameworks like LangChain, the system can analyze images and return JSON-formatted outputs. This approach can be adapted for fashion tech packs by feeding images of patterns or sketches and prompting the model to output key-value pairs like garment type, dimensions, and construction details. The workflow also incorporates OCR for improved accuracy with technical drawings containing measurement annotations.

Beyond commercial models, the research community has developed open models combining vision and language understanding. InstructBLIP (Dai et al., 2023) builds on the BLIP-2 vision-language model and applies instruction tuning so that it can follow natural language instructions about images. The model’s instruction-aware feature extraction focuses on relevant image details based on given prompts, ideal for identifying specific garment components from

sketches.

Liu et al. (2023) developed LLaVA (Large Language and Vision Assistant), which combines a vision transformer with a language model fine-tuned on visual instruction data. The LLaVA architecture processes images through a vision encoder (typically CLIP ViT-L/14) which extracts visual features. These features are then projected to the language model’s embedding space using a trainable projection layer. The combined embeddings are then processed by a language model (Vicuna-13B in the original implementation) that generates text responses based on both the visual input and text prompts. Figure 2 illustrates this process:

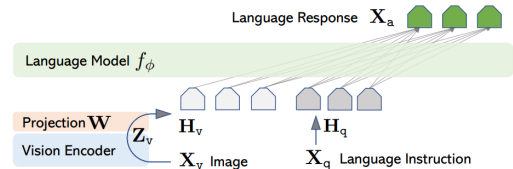


Figure 2: LLaVA architecture showing image and instruction processing pipeline

LLaVA can analyze visual content and generate detailed text descriptions when given specific instructions, making it applicable for converting fashion sketches into

technical documentation. The open release of LLaVA provides a foundation to build a custom multimodal assistant that could look at a garment pattern image and output a measurement table or construction steps.

3 Methodology

3.1 Research Approach

Our methodological approach for this project has consisted of several key phases:

First, we conducted a comprehensive analysis of existing research in both reinforcement learning for layout optimization and multimodal models for technical documentation generation. This literature review helped us identify the most promising techniques and approaches for our specific application in fashion production.

We then evaluated different algorithms and models based on several criteria:

- Applicability to irregular pattern shapes common in fashion
- Ability to handle domain-specific constraints like fabric grain
- Computational efficiency for real-time use in design workflows
- Potential for integration with existing design software
- Open-source availability for adaptation and extension

Based on this evaluation, we determined that PPO (Proximal Policy Optimization) represents the most promising approach for pattern layout optimization due to its demonstrated effectiveness with irregular shapes and ability to balance multiple objectives in the reward function. For component recognition and tech pack generation, we identified LLaVA as a particularly promising candidate due to its open-source nature and strong performance in visual understanding tasks.

3.2 Exploration of LLaVA

We have conducted initial experiments with the LLaVA model to assess its potential for garment component recognition. These experiments involved presenting the model with images of clothing items and evaluating its ability to identify and label key components such as collars, sleeves, and cuffs.

The typical input-output process for our LLaVA experiments follows this pattern:

- **Input:** An image of a garment sketch along with an instruction prompt such as "Identify all components in this garment sketch and label them with standard fashion terminology."
- **Processing:** The image is processed through the vision encoder, projected to the language model's embedding space, and combined with the text prompt.
- **Output:** A structured response identifying components (e.g., "This sketch shows a hoodie with the following components: 1) Kangaroo pocket, 2) Drawstring hood, 3) Ribbed cuffs, 4) Hem band, 5) Front zipper closure").

Our preliminary tests showed that LLaVA can successfully recognize major garment elements, though with varying degrees of accuracy depending on image clarity and component complexity. The model demonstrated particular strengths in identifying structural elements but sometimes struggled with more subtle features like stitching details or fabric properties.

These explorations have provided valuable insights into the potential of multimodal models for our task while also highlighting areas where additional training or prompt engineering might be necessary. We are currently developing plans for more systematic evaluation of LLaVA's capabilities and potential fine-tuning approaches to improve its performance on fashion-specific tasks.

3.3 Pipeline Design

Through our research and preliminary explorations, we have designed a comprehensive pipeline for our AI-driven fashion production system. This pipeline represents our roadmap for implementation. The key components include:

1. **Sketch Processing:** Converting designer sketches to vector format suitable for component recognition and pattern generation. We are evaluating Adobe Illustrator’s built-in Image Trace functionality for this conversion.
2. **Component Recognition:** Using vision-language models like LLaVA to identify garment components from processed sketches. This will require development of appropriate prompting strategies and potentially fine-tuning on fashion-specific data.
3. **Tech Pack Generation:** Leveraging multimodal LLMs to generate structured technical documentation from recognized components and optimized patterns. This will require development of standardized formats and templates for consistent output.
4. **Pattern Layout Optimization:** Implementing a reinforcement learning environment using OpenAI Gym to optimize pattern placement while respecting fabric grain and manufacturing constraints. PPO has been identified as the most promising algorithm for this task.
5. **Workflow Integration:** Creating a seamless connection between these components, potentially using frameworks like ComfyUI to orchestrate the complete process from sketch to production-ready documentation. ComfyUI’s node-based workflow system would allow us to create specialized nodes for each pipeline component, with custom nodes for sketch pro-

cessing, component recognition, pattern optimization, and tech pack generation that can be visually connected in a graph-based interface.

This pipeline design is based on our analysis of the state-of-the-art and represents our current understanding of the optimal approach. It will guide our implementation efforts in the next phase of the project.

4 Initial Results

4.1 Sketch-to-Vector Exploration

We have explored different approaches for converting sketches to vector format, focusing primarily on Adobe Illustrator’s built-in Image Trace functionality. Our investigation found that creating a custom image trace solution would be too extensive for our project scope, so we’ve focused on optimizing the preprocessing workflow for Illustrator’s existing capabilities.

As shown in Figure 3, our custom preprocessing approach (middle image) performs better at retaining all edges compared to Adobe Illustrator’s built-in tool (right image), though it still has some issues with noise. The original sketch (left image) shows a dress shirt design with detailed collar and cuff elements that our method preserves more effectively.

Our preprocessing strategy for sketches includes:

- Image enhancement through contrast adjustment and noise reduction
- Line stabilization and refinement using a Gaussian blur followed by thresholding
- Background removal to isolate garment outlines
- Resolution standardization to ensure consistent tracing results

After this preprocessing, the enhanced sketches can be fed into Illustrator’s Image Trace with optimized parameters for fashion sketches, including adjustments to path



Figure 3: Comparison of sketch-to-vector conversion methods. Left: Original sketch; Middle: Our custom preprocessing and vectorization; Right: Adobe Illustrator’s built-in Image Trace.

fidelity, corner angle detection, and noise thresholds.

Our analysis suggests that while Illustrator’s Image Trace provides high-quality vector conversion with good detail preservation, proper preprocessing is essential for consistent results. This approach balances visual quality and processing flexibility while working within our project constraints.

4.2 LLaVA Testing

Our preliminary tests with LLaVA for garment component recognition have shown promising potential. By presenting the model with images of clothing items and appropriate prompts, we observed that it can identify major structural elements such as collars, sleeves, bodice areas, and cuffs.

However, these tests also revealed limitations in the model’s current capabilities. LLaVA sometimes struggled with more technical terminology specific to fashion design and occasionally misidentified components with similar visual features. These observations suggest that domain-specific fine-tuning or carefully designed prompting strategies will be necessary to achieve reliable performance for our specific application.

While these results are encouraging, we have not yet developed quantitative metrics for evaluation, making it difficult to systematically assess performance or compare different approaches. Developing such metrics

will be an important next step in our research.

5 Research Pipeline and Future Work

Based on our research and initial explorations, we have refined our research pipeline:

1. Designer creates sketch in Adobe Illustrator
2. Illustrator’s Image Trace converts sketch to vector after preprocessing
3. Vector file exported to custom processing module
4. LLaVA identifies garment components from vector data
5. PPO algorithm optimizes pattern layout with fabric constraints
6. Optimized patterns passed to measurement extraction module
7. LLaVA generates tech pack specifications from components
8. ComfyUI orchestrates workflow with custom nodes for each stage

Our key research priorities for the next phase include:

- Investigating Adobe ExtendScript to automate vector export with component metadata

- Developing SVG parsing systems to extract meaningful garment features from Illustrator output
- Testing LLaVA performance on vector versus raster pattern recognition accuracy
- Creating specialized fashion domain datasets for fine-tuning LLaVA on garment component recognition
- Exploring penalty functions for fabric grain constraints in PPO optimization
- Designing JSON schema for standardized garment component specification
- Developing custom ComfyUI nodes for connecting optimization results to tech pack generation
- Researching effective reward structures balancing material usage and manufacturability
- Testing various PPO hyperparameter configurations for pattern nesting efficiency

Additionally, we plan to fork the LLaVA library to create a fashion-specific version tailored for tech pack generation. This will involve fine-tuning on garment sketches and corresponding technical specifications, developing specialized prompting techniques, and creating evaluation metrics for tech pack accuracy and completeness.

5.1 CLO3D Complementary Positioning

We’ve also examined CLO3D, a leading 3D fashion design software, to understand how our project can complement existing industry tools. CLO3D excels at creating realistic 3D visualizations of garments from 2D patterns but lacks capabilities in two key areas where our project focuses:

- **Sketch-to-Pattern Conversion:** CLO3D requires designers to create

pattern pieces manually or import them from specialized pattern-making software. Our system aims to bridge this gap by enabling automatic component recognition from design sketches.

- **Zero-Waste Optimization:** While CLO3D allows for virtual fitting and simulation, it doesn’t optimize pattern layouts for material waste reduction. Our reinforcement learning approach directly addresses this sustainability gap in the current workflow.

By positioning our system as a sustainability-focused add-on that could work alongside CLO3D, we can leverage the industry adoption of existing tools while adding critical functionality for waste reduction. We envision a workflow where designers could use our system to generate optimized pattern layouts, export them to CLO3D for visualization and fitting simulation, and then proceed to production with significantly reduced material waste.

To achieve this complementary positioning, we’ll design our API to be compatible with CLO3D’s file formats and create a simple interface that demonstrates the material and cost savings achieved through our optimization approach. This positioning allows us to focus on our core innovation—pattern layout optimization—while integrating with existing industry workflows.

6 Summary and Conclusion

In this mid-report, we have completed the following tasks:

- Conducted a comprehensive literature review of relevant techniques in reinforcement learning for nesting problems and multimodal models for documentation generation
- Compared different methodologies and algorithms to determine the most promising approaches

- Designed a comprehensive pipeline connecting design, optimization, and documentation
- Conducted preliminary explorations with LLaVA for component recognition
- Developed a detailed roadmap for implementation and evaluation
- Analyzed CLO3D to identify complementary positioning opportunities

The future work of this project will focus on:

- Implementing the sketch preprocessing and vector conversion pipeline
- Developing and testing the component recognition module using LLaVA
- Creating the reinforcement learning environment for pattern optimization
- Implementing the tech pack generation system
- Integrating these components into a cohesive workflow using ComfyUI
- Evaluating the system's performance on diverse garment designs
- Developing a simple interface that demonstrates optimization benefits

Through these efforts, we aim to create a comprehensive AI-driven system that addresses both environmental concerns through material waste reduction and efficiency challenges through automated documentation generation.

7 References

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